Automotive Radar Missing Dimension Reconstruction from Motion

Chun-Yu Hou[1], Chieh-Chih Wang[2], Wen-Chieh Lin[3]

***Abstract*— Automotive radars have been reliably used in most assisted and autonomous driving systems due to their robustness to extreme weather conditions. With radial velocity measurements from automotive radars, moving targets such as cars, trucks, and buses can be tracked robustly. However, due to the lack of elevation angles in measurements from automotive radars, stationary targets at different heights, such as maintenance holes and bridges, cannot be distinguished. Most autonomous systems rely on sensor fusion or ignore stationary targets to tackle the problem of missing the elevation angle dimension, which derives safety issues. We propose a simple yet effective approach to estimate the elevation angles of stationary targets from relative velocity and radial velocity measurements from an automotive radar. In contrast to structure from motion in computer vision, we utilize the instantaneous velocity generated from the motion of the ego vehicle. The radial velocity of each target is the projection of relative velocity onto the radial direction from radar to target. The radial velocity of each target can be inferred given the target's azimuth, elevation angle, and relative velocity. Accordingly, the elevation angle of each stationary target can be uniquely calculated given the velocity of radar and the target's azimuth and radial velocity measurements. The radar's velocity is estimated with the existing radar odometry algorithm and IMU. The proposed method is verified with real-world data. We evaluate the system's performance with a pre-built point cloud map and a good localization module in a real-world scenario. The proposed elevation angle reconstruction can reach a 1.41-degree mean error and standard deviation of 0.6 degrees in elevation angle.**

***Index Terms*— Automotive radar, Autonomous driving, 3D radar perception**

I. Introduction

3D automotive radars provide measurements of distance, azimuth angle, and radial velocity and have widespread applications in advanced driver assistance systems (ADAS). The advantages that radars provide extended sensing range and penetrating ability and are resilient to adverse weather conditions make radars well suited for autonomous driving applications [1]–[5]. With radial velocity measurements from automotive radars, moving targets such as cars, trucks, and buses can be tracked robustly [6].

However, the lack of elevation angle measurements with automotive radars has long troubled the applicability of radar usages. As shown in Fig. 1, automotive radars cannot discriminate whether the detected target is from the bridge or the ground. The stationary targets at different heights, such as bridges, will not result in a crash, but systems that use radars only may raise a false alarm. Most autonomous systems rely on sensor fusion [7] or ignore stationary targets to deal with this problem, which could cause safety concerns. Automotive radars were found malfunctioning in certain scenarios, including false slowdown under bridges [8]. Some accidents occurred when the autopilot systems crashed into parked cars [9]. The motivation of this paper is to address this incapability of automotive radars to distinguish the elevation of received targets.

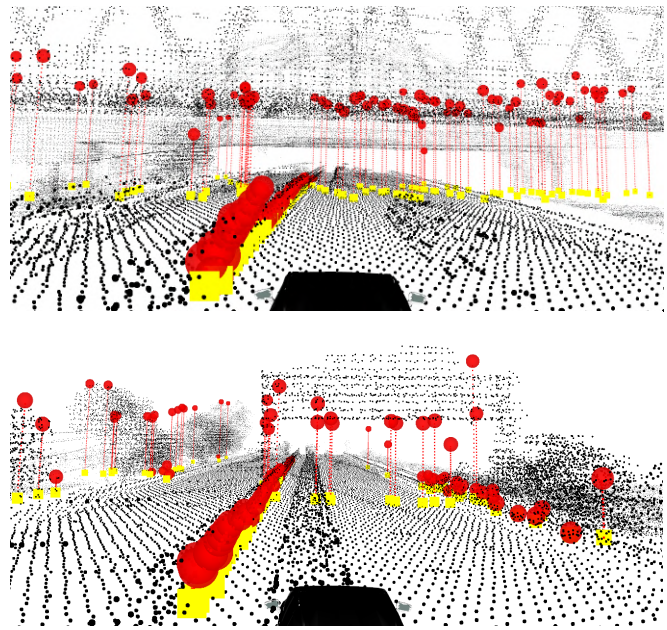

Fig. 1: An elevation angle reconstruction result in a highway scenario. Yellow squares are radar detected targets before elevation angle reconstruction. The reconstructed radar targets are shown in red spheres. Red dashed lines are the correspondences of the targets before and after reconstruction.

In this paper, we propose a simple yet effective method to reconstruct the missing dimension, elevation angle, of stationary targets from automotive radars. The core idea of this work is to estimate the elevation angles of stationary targets from the vector projection of relative velocity and radial velocity measurements. By definition, the radial velocity of every target is the projection of its relative velocity onto the relative direction from the radar sensor. The radial velocity of any target can be inferred if the target's azimuth, elevation angle, and relative velocity are known, as shown in Fig. 3. For stationary targets, the relative velocity equals

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the radar's 3D velocity in the opposite direction. Based on this constraint, the elevation angle can be calculated for every stationary target, given the 3D velocity of the radar and the target's azimuth and radial velocity measurements. By contrast, the elevation angles of moving targets cannot be inferred because the relative speeds to the radar sensor could be arbitrary. Fortunately, most moving targets could be detected and tracked with the existing motion segmentation and tracking algorithms without the elevation measurements.

Our proposed system for reconstructing elevation angles of stationary targets consists of three main components: 3D velocity estimation, stationary target detection, and elevation angle reconstruction. For the 3D velocity estimation, the 2D velocity and vertical velocity are estimated separately. The 2D radar velocity is estimated by using an extended Kalman filter to fuse the radar odometry [10] and IMU measurements to provide an accurate and robust estimation. The vertical velocity is calculated from the angular velocity measurement of IMU. With the estimated radar velocity, stationary target detection can be performed based on the radial velocity measurement of the radar. Accordingly, the elevation angles of stationary targets can be calculated with the above-mentioned relationship between velocities and spatial position.

Two challenges are encountered and overcome when we implement the proposed system in real-world settings. Firstly, we find that the accuracy of the radar's 2D velocity estimation using the general approach is affected when the radar is getting close to the bridge. The velocity estimation fails when the radar perceives a lot of targets at high elevations because the algorithm implicitly assumes that most perceived targets are stationary at radar height. This assumption is no longer valid when the number of targets at high elevations is significantly larger than the number of targets at radar height. As a result, the velocity estimation can be inaccurate, leading to errors in the elevation angle reconstruction. To address this problem, we utilize radar odometry and refine the estimated velocity to improve the accuracy and robustness of 2D velocity estimation. Secondly, although Kellner *et al.*'s algorithm [11] can detect stationary targets, we find that stationary targets at high elevations would be misclassified as moving targets. The radial velocity measurement of a stationary target at a high elevation is much smaller than that of a stationary target at radar height at the same azimuth, so it would be considered a moving target in the algorithm. To address this second challenge, we propose a stationary target detection method with a threshold varying with azimuth calculated from the radar's vertical field of view and the estimated radar's velocity.

In addition to the challenges discussed above, there are also limitations to our proposed method. Specifically, the accuracy of the reconstructed elevation angles is affected by two factors: the quality of the velocity estimation and the resolution of the radial velocity measurements from the radar. While we have developed strategies to mitigate errors arising from the velocity estimation, the resolution of the radial velocity measurements cannot be improved beyond the inherent limitations of the current automotive radar system. Therefore, the accuracy of the elevation angle reconstruction may be limited by the resolution of the radar's radial velocity measurements. Despite this limitation, we believe that our method represents a significant step forward in addressing the challenges of elevation angle reconstruction using automotive radars.

The proposed method is verified in a real-world scene. A pre-built LiDAR map with a good localization result is used to evaluate the system's performance. The proposed method reaches a 1.41-degree mean error and standard deviation of 0.6 degrees in elevation angle. Furthermore, we also tested a 4D radar system alongside our proposed method to establish a benchmark for comparison. Despite not outperforming the 4D radar in the current results, our method still holds considerable value. It provides a viable solution for elevation angle reconstruction using the existing automotive radar systems, which are more widely available and cost-effective compared to 4D radar technology. Our approach offers a practical and feasible option for enhancing the perception capabilities of standard automotive radar systems.

II. Related Work

The application of radar sensors has been widespread, with their performance improving significantly in recent years. More and more autonomous driving tasks can be solved with radar sensors. In this section, the applications of radar sensors that our method can help are discussed.

A. Radar Ego-motion Estimation

Motion estimation is crucial for autonomous driving systems. Kellner *et al.* [11] presented a robust algorithm to instantly estimate the velocity and yaw rate of the ego-vehicle with a single Doppler radar. Moving targets and clutter are considered outliers and can be extracted by RANSAC [12]. However, the heights of stationary targets were not considered. A stationary target at non-zero height will have a smaller radial velocity measurement than one at zero height. Their algorithm considered this effect derived from heights as sensor uncertainty. We draw inspiration from the above method to estimate the elevation angles of stationary targets from the radial velocity and ego velocity.

B. Radar Odometry and Localization

Kung *et al.* [10] proposed a normal distribution transform (NDT) based radar odometry achieving centimeter-level accuracy as lidar odometry which can be used on automotive and scanning radar. Ward *et al.* [13] propose a two-stage pipeline for radar localization with an EKF-based frame work integrated with ICP. DC-Loc [5] proposed a new radar-based metric localization framework that can obtain more accurate pose estimation by restoring the Doppler distortion. While the motion estimation and localization methods above can provide good results, they still lack full three-dimensional results. The main reason is that automotive radars only provide two-dimensional geometric measurements, missing elevation information. Our method has the potential to help 6-DoF

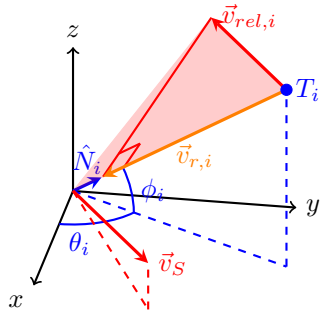


Fig. 2: The notation used in the paper. All are represented in the radar frame. The blue point is the ith target's position (T_i) and $\hat{N}_i$ is the unit vector pointing from the radar to the target. The transparent red triangle represents the radial velocity ($\vec{v}_{r,i}$) is the relative velocity's ($\vec{v}_{rel,i}$) projection on to the direction from the radar to the target.

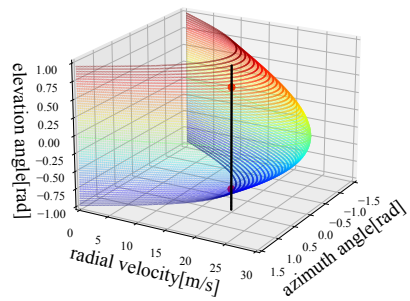


Fig. 3: The visualization of the distribution of radial velocity on azimuths and elevation angles. The black line intersects the surface at two points, which means that given a pair of radial velocities and azimuth angles, there are two possible solutions.

motion estimation and localization with automotive radar. By providing full three-dimensional information, our method could improve the safety and efficiency of autonomous driving systems and other applications that rely on motion estimation and localization.

III. Elevation Angle Reconstruction

This section is to derive the equations for reconstructing the elevation angle (ϕ_i) of the stationary target with automotive radar measurements ($|\vec{v}_{r,i}|, \theta_i$) and the velocity of radar ($\vec{v}_S$). Additionally, we observe the uncertainty of radial velocity measurement in the real-world experiment. The influence of radial velocity measurement uncertainty is discussed.

We first define the notation used throughout the paper. As shown in Fig. 2, the ith target's (T_i) position in the radar frame in spherical coordinate is denoted as radial distance, azimuth, and elevation angle: $[r_i, \theta_i, \phi_i]_{spherical}$ and can be transformed into Cartesian's space: $[r_i \cos\theta_i \cos\phi_i, r_i \sin\theta_i \cos\phi_i, r_i \sin\phi_i]_{cartesian}$. The radial velocity and relative velocity of the ith target relative to radar in the radar frame is denoted as $\vec{v}_{r,i}$ and $\vec{v}_{rel,i}$. The unit vector $\hat{N}_i = [\cos\theta_i \cos\phi_i, \sin\theta_i \cos\phi_i, \sin\phi_i]_{cartesian}$ denotes the unit vector pointing from radar to the ith target in the radar frame. The radar's velocity in the radar frame is denoted as $\vec{v}_S = [v_{S,x}, v_{S,y}, v_{S,z}]_{cartesian}$ and is assumed to be known in this section. Estimation of the radar's velocity and stationary target detection are covered in Section IV.

A. Velocity Projection

The main idea is to estimate the elevation angle from relative velocity and radial velocity measurements. By definition, the radial velocity of the ith target ($\vec{v}_{r,i}$) is the vector projection of the relative velocity ($\vec{v}_{rel,i}$) onto the direction ($\hat{N}_i$) from the radar to the ith target shown as the colored area in Fig. 2. For stationary targets, the relative velocities between the radar and the targets are the radar's velocities in the opposite direction. Thus, we can find the correspondences between the target's positions and the velocity of the radar:

$$|\vec{v}_{r,i}| = \vec{v}_{rel,i} \cdot (-\hat{N}_i) \tag{1}$$

$$\vec{v}_{rel,i} = -\vec{v}_S, \forall T_i \in \text{stationary target} \tag{2}$$

$$\begin{aligned} |\vec{v}_{r,i}| &= \vec{v}_S \cdot \hat{N}_i \\ &= \begin{bmatrix} v_{S,x} \\ v_{S,y} \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix} \cdot \cos\phi_i + v_{S,z} \cdot \sin\phi_i \end{aligned} \tag{3}$$

Please note that the motion of the ego-vehicle is necessary to compute the elevation angles. All relative velocities between the radar and stationary targets are zero without ego motion.

B. Radial Velocity Distribution and Application for 4D Radar

To depict the relationship between the target's radial velocity, azimuth, and elevation angle more clearly, the distribution of the radial velocity on azimuth and elevation angle is shown in Fig. 3. Given a non-zero $\vec{v}_S$, a corresponding distribution of radial velocities, azimuths, and elevation angles can be inferred.

In addition, this can be used to estimate the 3D velocity of the 4D radar and perform stationary target detection directly. Given θ, ϕ as measurements in 4D radar, $\vec{v}_S$ can be estimated using the same notion of [11] with RANSAC and least squares based on (3).

C. Elevation Angle Reconstruction

Given the velocity of the radar ($\vec{v_s}$) and $|\vec{v}_{r,i}|, \theta_i$ measurements from radar, the elevation angles of stationary targets can be calculated from (3):

$$\begin{aligned} \phi_i = \pm \cos^{-1}\Bigg(\frac{|\vec{v}_{r,i}|}{\sqrt{\left(\begin{bmatrix} v_{S,x} \\ v_{S,y} \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix}\right)^2 + v_{S,z}^2}}\Bigg) \\ + \tan^{-1}\Bigg(\frac{v_{S,z}}{\begin{bmatrix} v_{S,x} \\ v_{S,y} \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix}}\Bigg) \end{aligned} \tag{4}$$

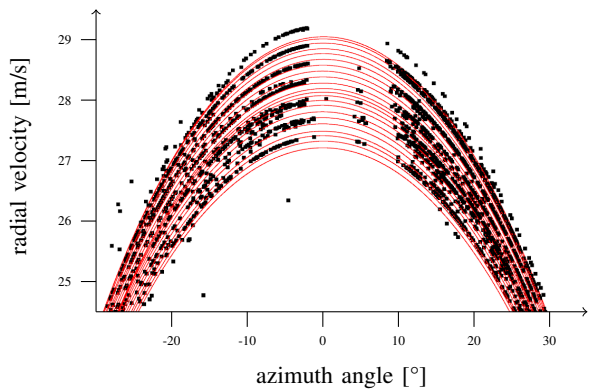


Fig. 4: Stacking a period of time of the velocity profile during speed change. Black points are all received targets over time. Red lines represent the radar's velocity calculated from localization results. It is observable that there are gaps between the curves composed by black points.

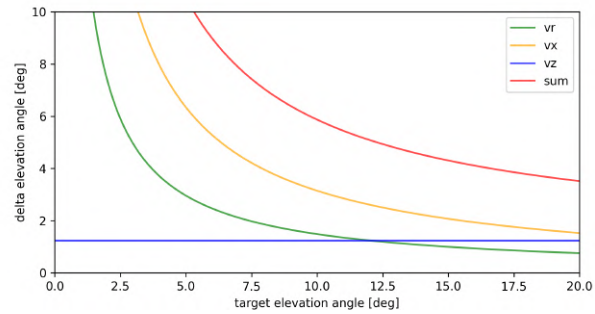


Fig. 5: The influence of each input to elevation angle based on the target's elevation angle.

However, the sign of the arccosine component remains undetermined. As shown in Fig. 3, given azimuth angle and radial velocity, the black line intersects the surface at two possible elevation angles. To address this issue, we assume all targets' elevation angles are not smaller than zero. In other words, all targets are assumed to be not lower than the xy plane in the radar frame. Since the arccosine component dominates the elevation angle when the car's speed is high, we can eliminate the negative solutions of elevation angles. The assumption is made since all targets are above the radar plane in most driving scenarios. In our experiments, the radar is mounted on the front bumper of the vehicle, 0.8m height from the ground, which is sufficiently close to the ground. With this assumption, the negative part is ignored and we can determine the target's elevation angle. While this assumption may not hold in all scenarios, it is a reasonable approximation in most driving situations where targets are not located on the ground. It should be noted, however, that in scenarios where targets are located on the ground, this assumption may not hold, and the resulting elevation angles may be inaccurate. It is the limitation of using a single automotive radar to reconstruct elevation angles. A straightforward method is to use another automotive radar mounting on a higher location on the car. Associating the targets between two radars could easily solve this problem. With the data from both radars, it becomes possible to determine the target's elevation accurately.

D. Uncertainties of Elevation Angle Reconstruction

To investigate and compare the impact of errors on all inputs of the elevation angle reconstruction method, the Taylor series expansion of (4) is used. We take the first-order partial derivative of the elevation angle reconstruction equation with respect to each input variable to investigate and compare the influence of errors on the method. Our observation indicated that low resolution in radial velocity is pivotal for the reconstruction of the elevation angle, especially when there is a change in the ego vehicle's speed. This resolution can be clearly seen in Fig. 4, where noticeable gaps in radial velocity emerge upon aggregating measurements over a specific time period. To explain further, we employed the Continental ARS408-21 radar for our study. The radial velocity in this system is determined by projecting the longitudinal and lateral velocity readings, delivered by the radar, onto the radial direction. Both of these velocities are output at a resolution of 0.25 m/s. Consequently, this resolution corresponds to the gaps we observe in Fig. 4, with the maximum gap being 0.25 m/s. In a highway scenario, with the ego vehicle cruising at 100 km/h, we focus on the target ahead of the car. Three errors come into play for different variables: The errors of $v_{r,i}$ is considered as 0.125 m/s, which is half of its resolution. As for $v_{S,x}$, we consider the radar odometry's error, which is measured at 0.269 m/s. Additionally, for $v_{S,z}$, its error stands at 0.6 m/s, derived from our experiment using the IMU. Given the car's high forward speed and the relatively small error in the azimuth angle, we can safely ignore $v_{S,y}$ and θ_i in our current analysis. As shown in Fig. 5, the contributed error of v_r and v_x increase significantly as the target elevation angle gets close to zero. The $\cos\phi$ term in (3) makes the radial velocity less sensitive in small elevation angles because of the small slope of cosine near zero. Thus, it's important to note that our proposed method may be subject to higher errors when dealing with targets possessing smaller elevation angles.

IV. 3D Velocity Estimation and Stationary Target Detection

This section is to estimate $\vec{v}_S$ and perform stationary target detection, which is essential for reconstructing the elevation angles of stationary targets. The x, y components and the z component of $\vec{v}_S$ are estimated separately, then the stationary target detection is performed based on the estimated $\vec{v}_S$.

A. 2D Velocity Estimation

To estimate the 2D velocity $[v_{S,x}, v_{S,y}]$ of the radar, we utilize a normal distribution transform-based radar odometry [10] and an extended Kalman Filter incorporating IMU.

1) Radar Odometry: The normal distribution transform-based radar odometry method consists of a probabilistic submap module, an NDT module, and an NDT matching module. The process of constructing the probabilistic radar submap involves integrating multiple radar scans while considering the measurement uncertainty and the ego motion.

It involves deploying the ego-velocity estimation, which is calculated using radial velocity measurements [14] to combine multiple radar scans. Then, the normal distribution transform module converts the combined radar scan into a set of the local normal distribution functions taking uncertainties into account, which is the probabilistic NDT. After the probabilistic NDT is generated, the weighted Point-to-Distribution (P2D) NDT matching is used to align two radar scans, and the displacement of the car can be obtained.

An extended Kalman Filter, a recursive state estimation algorithm widely used in sensor fusion applications, is deployed in this scenario to seamlessly fuse and integrate the estimation of radar odometry and the measurements obtained from an Inertial Measurement Unit (IMU). By leveraging the complementary strengths of these two distinct sensing modalities, the extended Kalman Filter ensures a robust estimation of the ego velocity, accounting for potential uncertainties and enhancing the overall accuracy and reliability of the velocity estimation process. The x and y position, as well as the yaw angle, are obtained from radar odometry, while the x and y accelerations and the yaw rate are derived from the IMU. By integrating these different sources of information, the extended Kalman Filter achieves a robust estimation of ego velocity, improving accuracy and reliability.

2) Refine 2D Velocity: To ensure the estimated 2D velocity conforms to the velocity profile of the current radar scan. The velocity obtained from the EKF is used as an initial guess, and the velocity is recalculated with the closest 50% of targets in radial velocity-azimuth space using the RANSAC and least square proposed in [11]. This ensures the estimated velocity conforms to the radial velocity's distribution over the azimuth. On the other hand, compared to applying RANSAC directly as used in [11], using radar odometry as an initial guess and refining is more accurate in environments with many targets in high elevation. We observe that the velocity is often underestimated under bridges in experiments because the radar received most of the targets from the bridge, as shown in Fig. 6. As the nature of RANSAC is fitting to the major group of targets when the radar receives most of the targets from the bridge, which is the dotted line in Fig. 6 right. In comparison, the proposed method of using the EKF and refining (red line) improves the robustness of 2D velocity estimation and maintains the accordance of the estimated 2D velocity with the velocity profile of the current radar scan. By leveraging the Extended Kalman Filter (EKF) to obtain an initial velocity estimate and then refining it using the RANSAC (Random Sample Consensus) algorithm in conjunction with least squares, we can achieve better accuracy and robustness in velocity estimation, especially in scenarios with numerous targets in high elevation.

B. Vertical Velocity

As stated in Section III-D, the vertical velocity $v_{S,z}$ impacts the elevation angle reconstruction result. It is essential to reconstruct the elevation angle with the full 3D velocity of radar. Therefore, IMU is integrated into the reconstruction system in which the vertical velocity of the radar is estimated.

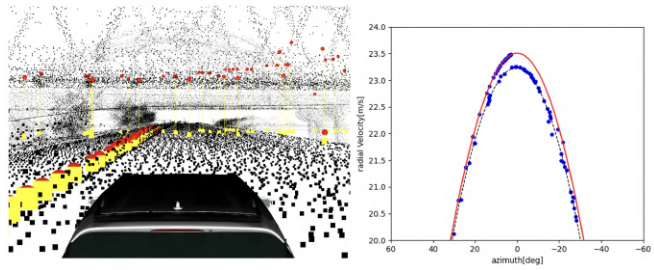
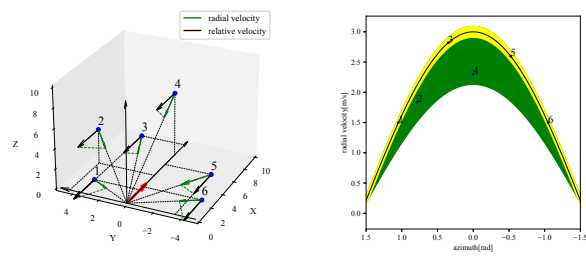


Fig. 6: An example of 2D velocity estimation using RANSAC influenced by the bridge. The red line in the right represents the velocity estimated by the proposed method; the dotted line represents the velocity estimated using RANSAC

The vertical velocity of the radar sensor is assumed to be derived from the bumpy road. We consider the motion caused by a bumpy road to be a pitch rotation. Thus, the vertical velocity of radar v_z is calculated from the angular velocity measurement $\vec{\Omega}$. Only the vertical velocity caused by rotation is extracted:

$$v_{S,z} = D_{radar,IMU} \cdot \vec{\Omega} \cdot \hat{z}_{radar} \tag{5}$$

where $D_{radar,IMU}$ is the distance from the radar to IMU; $\vec{\Omega}$ is the angular velocity expressed via Euler angle; $\hat{z}_{radar}$ is the vertical direction unit vector of the radar frame in the IMU frame.



(a) A scene for demonstration (b) Velocity profile

Fig. 7: A toy problem demonstrating the elevation angle affects the radial velocity in the velocity profile. Fig.7a demonstrate the scene where the radar sensor is placed at (0,0,0). The red arrow denotes the velocity of the radar sensor, while the blue and green arrows denote the relative velocity and estimated radial velocity of each target, respectively. Fig.7b is the velocity azimuth plot of the scene. The line represents the radar's 2D velocity and should intersect with all stationary targets at zero height. Targets 1,3,5,6 are the desired targets for radar's 2D velocity estimation. The corridor threshold for 2D velocity estimation is the yellow area. Target 2,4 are stationary targets at high elevations, which should be kept in stationary target detection. The corridor threshold for stationary target detection is the green area.

C. Stationary Target Detection

The typical approach to stationary target detection is using a constant corridor threshold on the radial velocity based on the velocity profile, which represents the distribution of the azimuth versus the radial velocity of stationary targets [11]. The velocity profile describes the distribution of the i-th stationary target's radial velocity $v_{r,i}$ over its azimuth angle θ_i where all stationary targets should confirm this distribution in a 2D flat scenario. A corridor threshold is applied to

classify stationary and non-stationary targets, as shown in the yellow area in Fig. 7b. However, in a non-flat scenario, for example, passing through a bridge, stationary targets at high elevations would be misclassified as non-stationary. As described in the previous section, the absolute value of the radial velocity measurement will decrease while the height of the stationary target at the same azimuth increases. Thus, when the decreased value in radial velocity exceeds the corridor threshold, the stationary target at high elevation will be misclassified as non-stationary, for instance, target 2 and 4 in Fig. 7. To mitigate this problem, an easy solution is to use a bigger corridor threshold to include those targets; however, this will result in misclassifying moving targets as stationary. Instead of a constant threshold value, we propose to adjust the threshold value based on the vertical field of view of the radar. This corridor threshold is determined according to the estimated $\vec{v}_S$ and radar sensor's vertical field of view (ϕ_{FOV}) from Eq. 3. It should be noted that the corridor threshold varies with the azimuth angle θ, which made the corridor threshold (v_{thr}) for stationary target detection (Fig. 7b green area) a function of θ:

$$v_{thr}(\theta) = \begin{bmatrix} v_{S,x} \\ v_{S,y} \end{bmatrix} \cdot \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \cdot \cos\phi_{FOV} + v_{S,z} \cdot \sin\phi_{FOV} \quad (6)$$

By dynamically adjusting the corridor threshold based on the radar's field of view, the proposed method optimizes the stationary target detection process and improves the efficiency of target classification in complex, non-flat scenarios such as passing through a bridge.

It should be noted that the corridor threshold is bigger, which may result in more moving targets misclassified as stationary targets at high elevations. Consider a highway driving scenario where the car speed is 100 km/h, we are focusing on the target in front of us, and the radar's vertical field of view is 20°. The value of the tolerance of radial velocity at zero azimuth ($|v_S| - v_{thr}(0)$) is about 7.7 km/h, which means a car moving at speed slower than 7.7 km/h would be misclassified as a stationary target at some elevation angle between zero and the vertical field of view. Though this is not a common case that would be encountered in highway driving scenarios, it still warrants attention as it could present potential hazards.

V. Experimental Results

The evaluation of the system was conducted using 5 Continental ARS 408-21 automotive radar sensors surrounding the vehicle, along with Xsens MTI-300 IMU. Only the front radar is used to perform elevation angle reconstruction; the other four radars are used in the radar odometry. The vehicle was also equipped with a 128-beam LiDAR (Velodyne VLS-128) that provided reliable localization with respect to the point cloud map. To ensure synchronization with the radar timestamps, the radar poses were interpolated from the LiDAR localization results. The evaluation involved a sequence of 204 radar frames, covering a distance of approximately 450 m and lasting for 15 s, in a highway scenario with the vehicle traveling at a speed of around 100 km/h. The test sequence included passing under two bridges (Fig. 8).

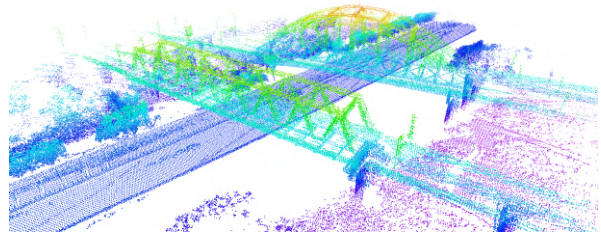


Fig. 8: The point cloud map of the two bridges in the test sequence.

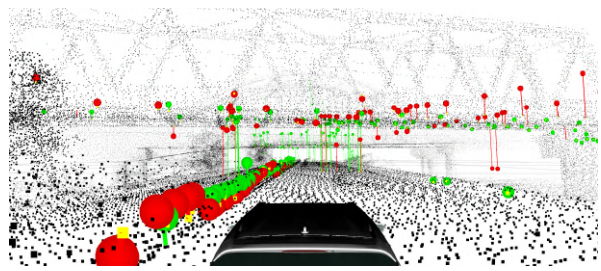


Fig. 9: Red points are the result of elevation angle reconstruction; green points are 4D radar measurements. Lines represent the correspondence established to the LiDAR map, respectively. Yellow squares are the corresponding map points.

Additionally, a Continental ARS 548 4D radar is tested together to provide a reference for the proposed method. To ensure a fair comparison, targets outside the 3D automotive radar's field of view (FOV) were removed, as the FOV of the 4D radar is larger. We then employed the stationary target detection method discussed in Section III-B to extract stationary targets.

We assess the reconstruction performance in real-world scenarios by establishing a pseudo correspondence with a pre-built point cloud map. Choosing a point cloud map for evaluation is because of the range and penetrating capability of radar. On the other hand, although moving targets are removed in the algorithm, we have carefully checked that there are no moving objects interfering with the reconstruction results. We establish a correspondence from the reconstructed radar target to the closest point in the same azimuth angle and range on the point cloud map, as shown in Fig. 9. A tolerance of 0.1° in the azimuth angle and 0.4m in the range is added to account for the uncertainty of these measurements. The angle between two corresponding points is calculated to provide an error term invariant to the target's distance.

TABLE I: Elevation Angle Error

	Mean Angle Error[°]	Standard Deviation
Elevation Angle Reconstruction	1.41	0.60
4D Radar	0.63	0.25

The elevation angle errors are shown in Table I. As shown in Fig. 10, the mean angle error is higher around 2m, which is the height of the traffic island, compared to other heights. This observation aligns with the analysis in Section

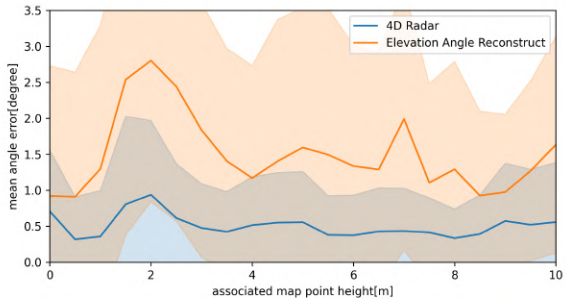


Fig. 10: The mean angle error of different heights. The colored area represents the standard deviation.

III-D, suggesting that errors tend to be higher for smaller elevation angle targets. The error is considered derived from the deviation in the radial velocity measurements. Taking the radial velocity uncertainty into account and transforming this uncertainty to elevation angle with (4), the transformed uncertainty covers where it should locate. As shown in Fig. 11, the uncertainty of the reconstructed target covers the point where it should locate. It is worth noticing that due to the gradient of arccosine at different value in (4), the uncertainty for targets at higher altitude are much smaller than those at low altitude (Fig. 11 below). Though not surpassing the 4D radar, our method remains valuable, offering a viable solution for elevation angle reconstruction using existing automotive radar systems.

The mean execution time of our proposed method is approximately 4.4 ms (with a standard deviation of 2.4), as measured over the experiment sequence (tested on Intel i7-7700K CPU, 4.5 GHz). The method's execution is dominated by the velocity estimation step because of the RANSAC algorithm in the refine velocity step. The execution time of the velocity estimation and stationary target detection step is approximately 3.5 ms. The time delay for the elevation angle reconstruction step is about 0.8 ms.

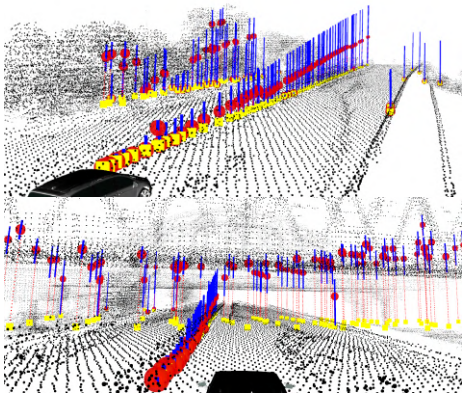


Fig. 11: Yellow squares represent radar targets before elevation angle reconstruction. The reconstructed radar targets are shown in red spheres. Blue lines are the uncertainty from radial velocity measurements transformed into 3D space.

VI. Conclusion and Future Work

We propose a simple yet effective approach to estimate the elevation angle of stationary targets detected by automotive radars from motion. With the method we propose, the elevation angle of stationary targets can be uniquely calculated from radial velocity, azimuth measurements, and radar velocity. Our experimental results show that the proposed method can reach a 1.41-degree mean error and standard deviation of 0.6 in elevation angle. We also point out a different perspective to inspect the velocity profile by introducing the elevation angle into the velocity profile. The current method is based on a single cycle of measurement; we would like to extend the method over time to deal with the measurement uncertainties.

References

[1] P.-C. Kung, C.-C. Wang, and W.-C. Lin, "Radar occupancy prediction with lidar supervision while preserving long-range sensing and penetrating capabilities," *RAL*, vol. 7, no. 2, pp. 2637–2643, 2022.

[2] Y. Zhou, L. Liu, H. Zhao, M. López-Benítez, L. Yu, and Y. Yue, "Towards deep radar perception for autonomous driving: Datasets, methods, and challenges," *Sensors*, vol. 22, no. 11, 2022.

[3] X. Dong, P. Wang, P. Zhang, and L. Liu, "Probabilistic oriented object detection in automotive radar," in *CVPR Workshops*, June 2020.

[4] E. Marti, M. A. de Miguel, F. Garcia, and J. Perez, "A review of sensor technologies for perception in automated driving," *IEEE Intelligent Transportation Systems Magazine*, vol. 11, no. 4, pp. 94–108, 2019.

[5] P. Gao, S. Zhang, W. Wang, and C. X. Lu, "Dc-loc: Accurate automotive radar based metric localization with explicit doppler compensation," in *ICRA*, 2022, pp. 4128–4134.

[6] A. Eltrass and M. Khalil, "Automotive radar system for multiple-vehicle detection and tracking in urban environments," *IET Intelligent Transport Systems*, vol. 12, no. 8, pp. 783–792, 2018. [Online]. Available: https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/iet-its.2017.0370

[7] Y. Long, D. Morris, X. Liu, M. Castro, P. Chakravarty, and P. Narayanan, "Radar-camera pixel depth association for depth completion," in *CVPR*, June 2021, pp. 12 507–12 516.

[8] A. Karpathy, "[cvpr'21 wad] keynote - andrej karpathy, tesla," https://youtu.be/g6bOwQdCJrc, June 2021.

[9] B. Templeton, "Tesla in taiwan crashes directly into overturned truck, ignores pedestrian, with autopilot on," https://www.forbes.com/sites/bradtempleton/2020/06/02/tesla-in-taiwan-crashes-directly-into-overturned-truck-ignores-pedestrian-with-autopilot-on/?sh=681c80fa58e5, June 2020.

[10] P.-C. Kung, C.-C. Wang, and W.-C. Lin, "A normal distribution transform-based radar odometry designed for scanning and automotive radars," in *ICRA*, 2021, pp. 14 417–14 423.

[11] D. Kellner, M. Barjenbruch, J. Klappstein, J. Dickmann, and K. Dietmayer, "Instantaneous ego-motion estimation using doppler radar," in *ITSC*, 2013, pp. 869–874.

[12] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, no. 6, p. 381–395, 1981.

[13] E. Ward and J. Folkesson, "Vehicle localization with low cost radar sensors," in *2016 IEEE IV*, 2016, pp. 864–870.

[14] D. Kellner, M. Barjenbruch, J. Klappstein, J. Dickmann, and K. Dietmayer, "Instantaneous ego-motion estimation using multiple doppler radars," in *2014 IEEE ICRA*, 2014, pp. 1592–1597.